

# Quibrantol — Stability Study Design

Protocol STAB-P-2025-43

## Analytical Procedures

Assay and degradation products are determined by the stability-indicating reversed-phase HPLC procedure described in the analytical methods dossier. Dissolution testing uses USP Apparatus II at 50 rpm. All procedures were validated in accordance with ICH Q2(R1) prior to protocol initiation.

## Storage Conditions

The primary (zone IVb condition) storage condition for this protocol is  $30^{\circ}\text{C} \pm 2^{\circ}\text{C}$  /  $75\% \text{ RH} \pm 5\% \text{ RH}$ . Accelerated studies at  $40^{\circ}\text{C} \pm 2^{\circ}\text{C}$  /  $75\% \text{ RH} \pm 5\% \text{ RH}$  were completed through six months during development and are not part of this protocol.

## Timepoints

| Interval (months) | 0 | 1 | 3 | 6 | 9 | 12 | 18 |
|-------------------|---|---|---|---|---|----|----|
| Long-term testing | X | X | X | X | X | X  | X  |

## Bracketing Justification

In accordance with ICH Q1D, bracketing is applied: the 10 mg and 40 mg strengths are tested, and the intermediate 25 mg strength is covered because all strengths are direct scale-equivalents of a common blend in the same container closure system.

## Stability Commitments

A post-approval commitment is made to enrol one commercial batch of Quibrantol per year into the ongoing stability program and to report confirmed out-of-specification results within 30 days.

## Protocol Deviations

The 6-month samples for batch 003 were held at ambient conditions for 12 hours during chamber maintenance before testing; trending showed no anomalous results.

## Acceptance Criteria

The stability acceptance criterion applied at each interval is — Water content: not more than 5.0% w/w. All other registered specification tests apply unchanged.

## **Sample Inventory and Pull Windows**

Sample inventory per condition and timepoint is defined in the pull schedule appendix. Scheduled pulls are performed within  $\pm 7$  days of the nominal date; pull records are maintained in the site stability management system.

## **Batch Selection**

The protocol covers three primary batches manufactured at commercial scale using the approved process. Batch genealogy, manufacturing dates, and packaging lots are recorded in the batch appendix. All batches were released against the approved specification prior to set-down.

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## **Container Closure and Orientation**

Samples are stored in the commercial container closure system. Bottles are stored upright; a subset of inverted samples is included at the initial and final timepoints to assess closure-liner interaction. Induction seal integrity is verified at each pull.